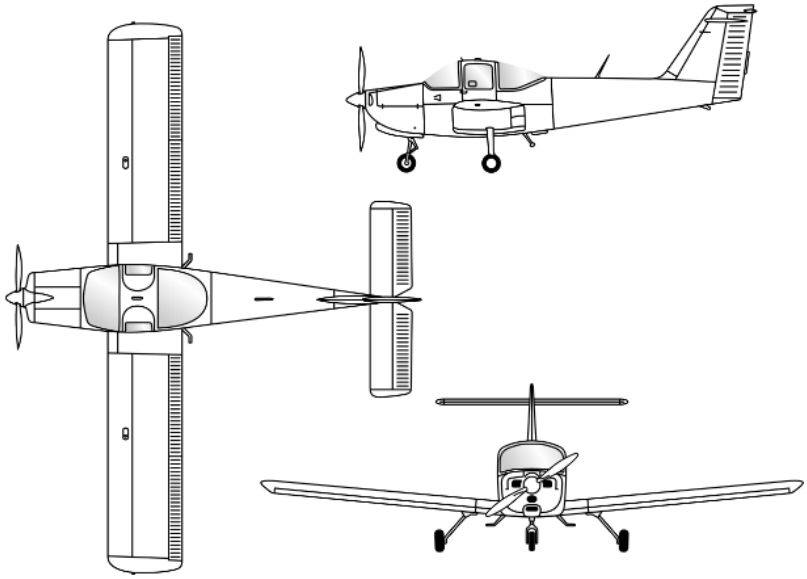
Introduction



This report presents the detailed measures and strategies adopted to design, build, and test an aircraft for the aircraft manufacturing course in the sixth semester of BAS. The aircraft manufacturing engineering course demands clear design, fabrication and manufacturing of the necessary parts, optional parts of an aircraft. The team decided to scale down the existing model of Piper PA-38 Tomahawk, which is a low-wing, tractor-motor aircraft with a T-tail empennage. A rectangular wing having AR 9.33, span 10.36 m with 1.11 m chord, with a fuselage of length 7.01 m and a width of 1 m was made. To be precise, this report outlines the preliminary and conceptual design for a 1:14 scale model of the Tomahawk, considering aerodynamic, structural, and propulsion aspects.

Preliminary Design

The first step to design was scaling down the actual prototype by a ratio of 1:14. All dimensions of the full-scale aircraft were divided by 14. The real Piper PA-38 Tomahawk has a fuselage length of 7.01 meters (701 cm), which results in a scaled-down length of 50.07 cm. The wingspan, originally 10.36 meters (1036 cm), scales down to 74.0 cm, while the wing area reduces from 11.52 square meters to 0.0589 square meters. The average wing chord, calculated as the ratio of wing area to wingspan, is reduced from 111 cm to 7.93 cm. The aircraft's height, originally 2.74 meters (274 cm), scales down to 19.57 cm. Despite these dimensional changes, the aspect ratio remains the same at 9.33, ensuring that the aerodynamic properties of the model closely resemble those of the full-scale aircraft. The dimensions are shown in the table below:

Scaling Considerations:

Full-Scale Dimensions:	1:14 Scale Dimensions:
Length: 7.01 m	Length: 0.50 m (50 cm)
Wingspan: 10.36 m	Wingspan: 0.74 m (74 cm)
Wing Area: 11.52 m^2	Wing Area: 0.59 m^2
Chord Length: 1.11 m (average)	Chord Length: 7.9 cm

Design Considerations

Materials Selection

- Fuselage: Plywood
- Wings: Plywood
- Tail: Plywood

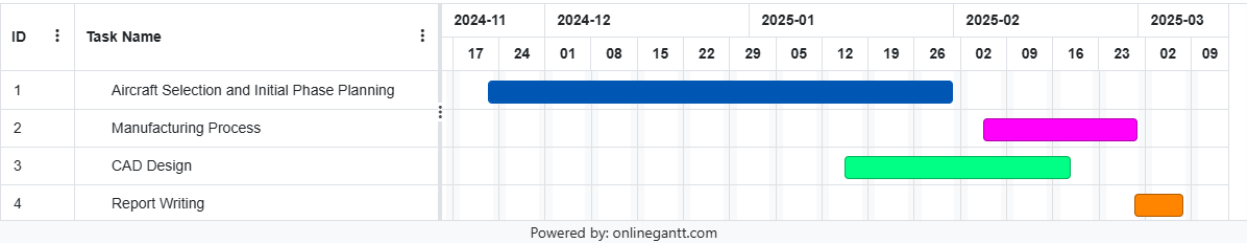
Structural and Aerodynamic Design

- Wing Loading Calculation:

Assuming a model weight of 400g and a wing area of 0.0589 m², the estimated wing loading is 6.79 kg/m².

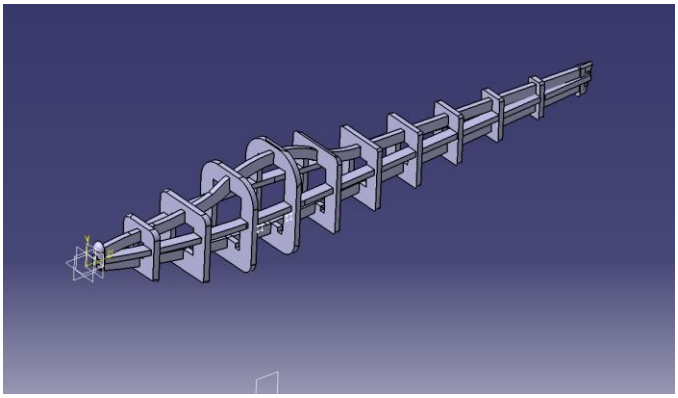
- Airfoil Selection: NASA GAW-1 for stability.
- Dihedral Angle: ~5° for improved roll stability.
- T-Tail Stability: Additional structural reinforcement required.
- CG (Center of Gravity): Positioned at ~25–30% of the Mean Aerodynamic Chord (MAC).

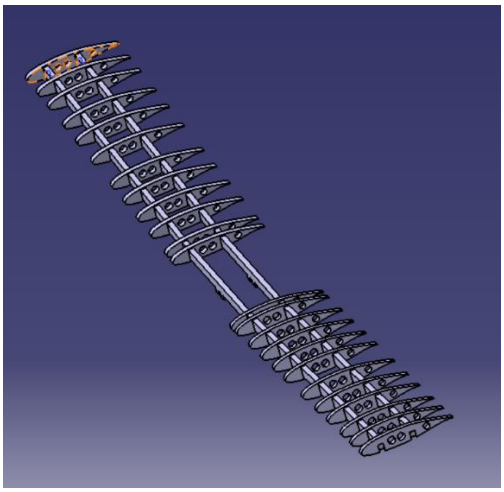
The Gantt chart for the project planning is given below:



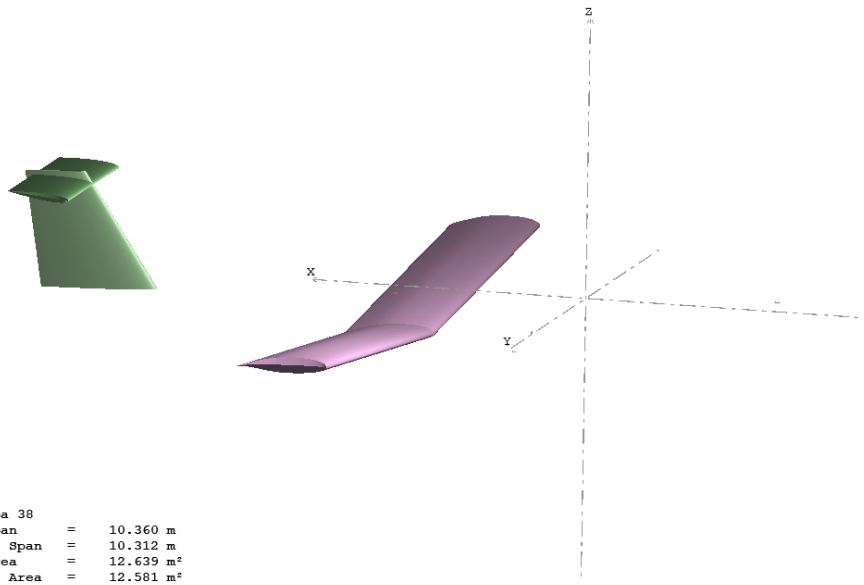
Detail Design

The wing design follows a careful balance between structural integrity and aerodynamic efficiency. The estimated wing loading, calculated as the total weight divided by the wing area, is approximately 6.79 kg/m² if the model weighs 400 grams. This value is appropriate for small-scale aircraft and ensures adequate lift generation. NASA GAW-1 airfoil is chosen due to its well-documented stability and performance in subsonic conditions. The wings will also incorporate a dihedral angle of approximately 5 degrees to improve roll stability. The design process was carried out in XFLR5. The team members developed CAD file for the allocated parts and the assembly was done at last. The assembled fuselage and wing is shown in the picture below:



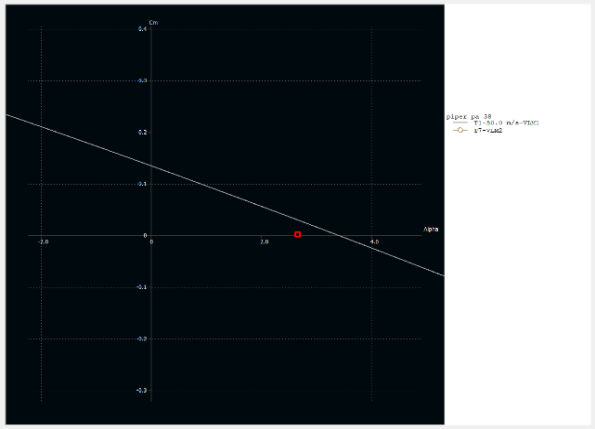
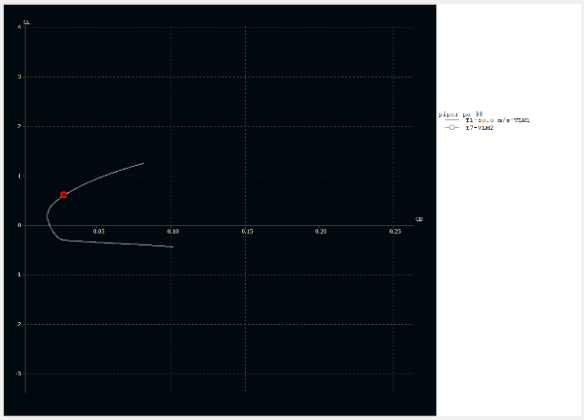
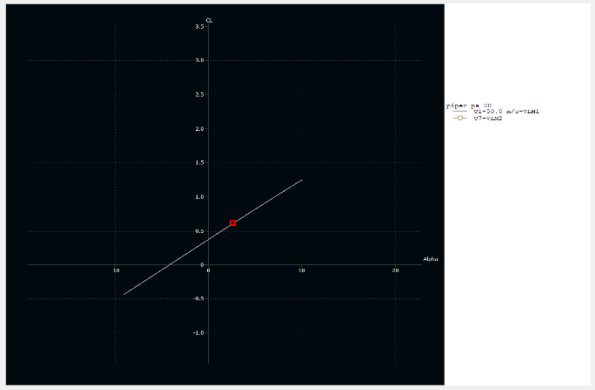
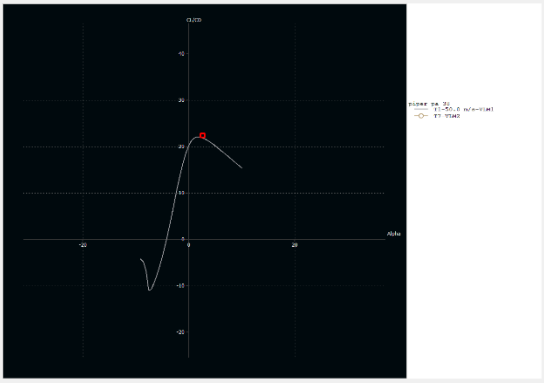
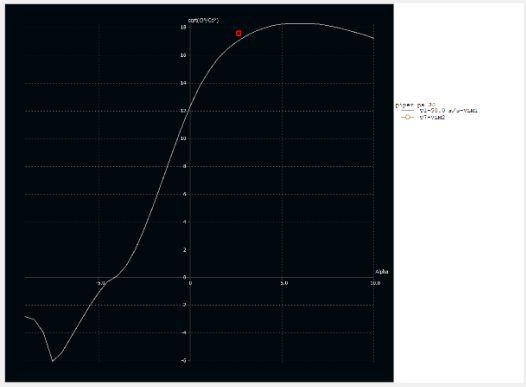


All of the initial phase tests were carried out in XFLR5 with appropriate analysis for suitable airfoil and other batch analysis in suitable Reynold's number.

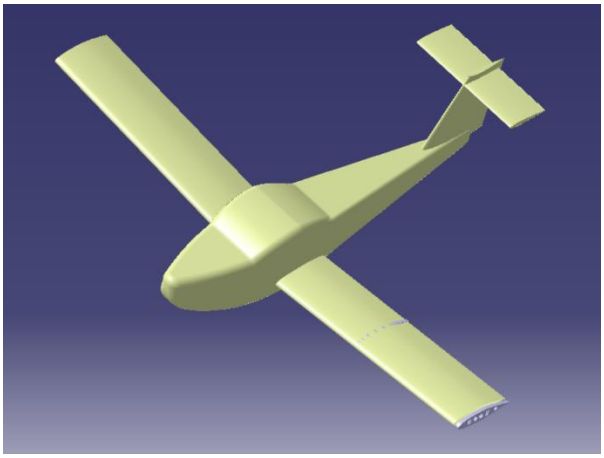



```
piper pa 38
Wing Span      =    10.360 m
xyProj. Span   =    10.312 m
Wing Area      =    12.639 m²
xyProj. Area   =    12.581 m²
Plane Mass     =   757.500 kg
Wing Load      =    60.210 kg/m²
Tail Volume    =     0.583
Root Chord     =     1.220 m
MAC            =     1.220 m
TipTwist       =     0.000°
Aspect Ratio   =     8.492
Taper Ratio    =     1.000
Root-Tip Sweep =     0.000°
XNP = d(XCp.Cl)/dCl =     0.000 m
Mesh elements  =   809
```

```
          V =   46.00 m/s
      Alpha =    2.653°
       Beta =    0.000°
         CL =    0.615
         CD =    0.027
 Efficiency =    0.754
      CL/CD =   22.383
         Cm =    0.002
         Cl =    0.000
         Cn =    0.000
       X_NP =    2.704 m
       X_CP =    2.194 m
       X_CG =    2.189 m
```

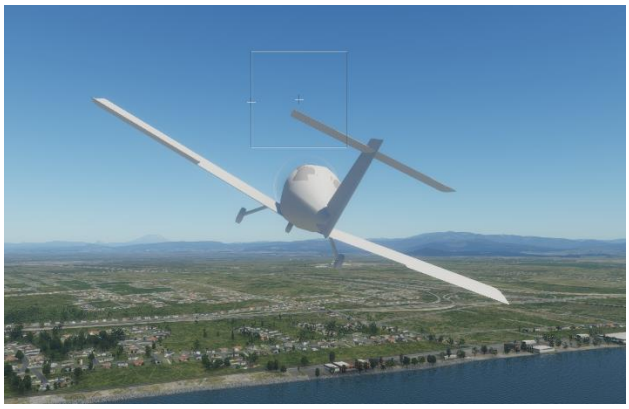






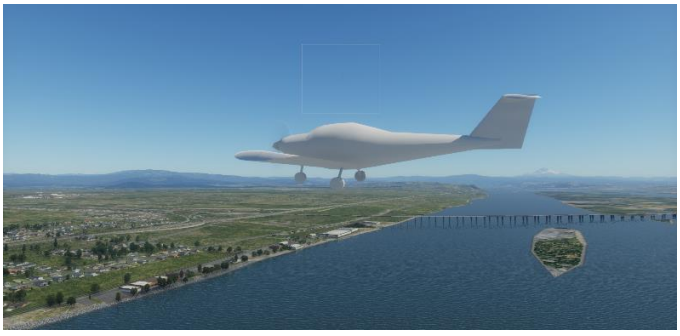
For the tail section, the horizontal stabilizer spans approximately 12 cm, while the vertical stabilizer stands around 6 cm in height. Special attention is given to reinforcing the T-tail mount, as the high placement of the stabilizer increases stress on the fuselage attachment point. The final assembled aircraft looked like the figure shown below.

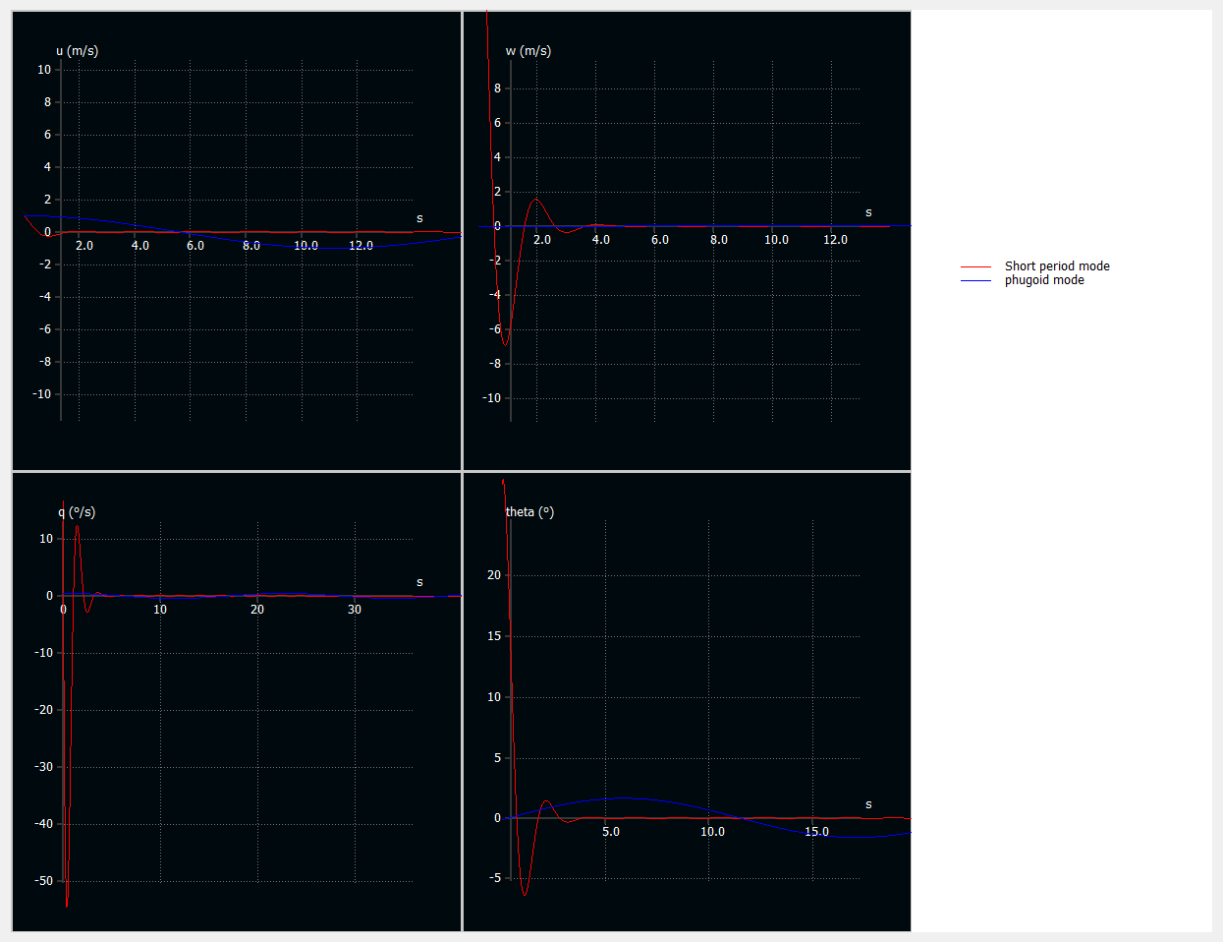


The further simulation was carried out in XPlane to check the flyability of the aircraft and the result was positive.

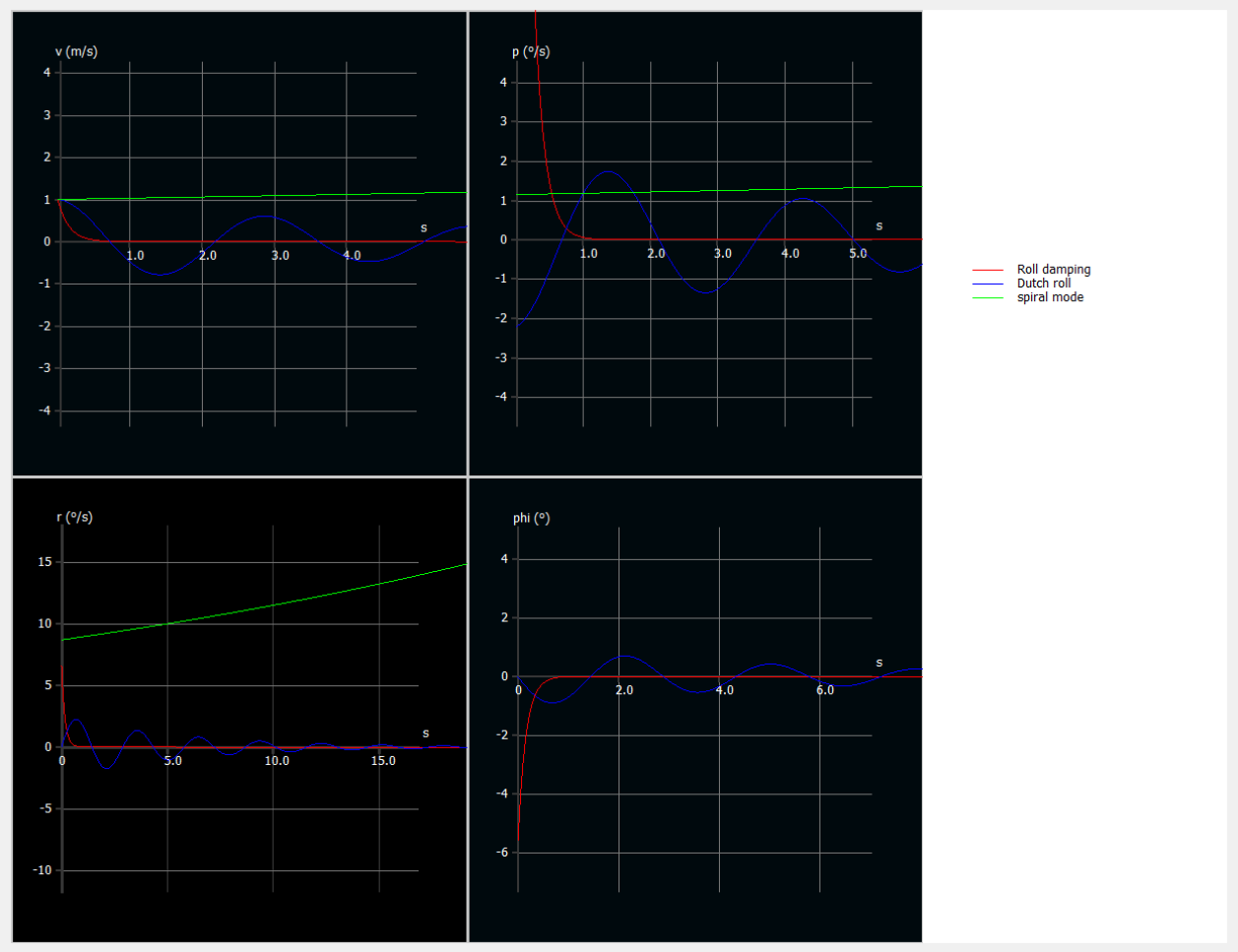






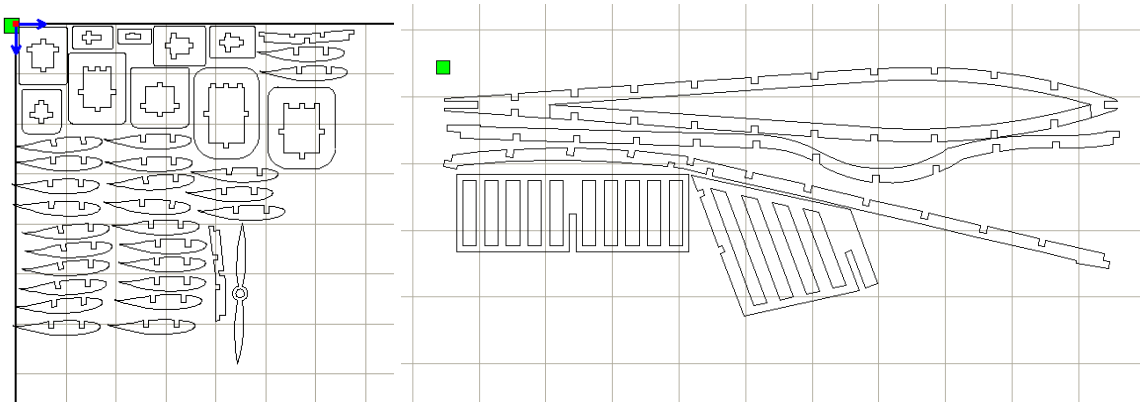


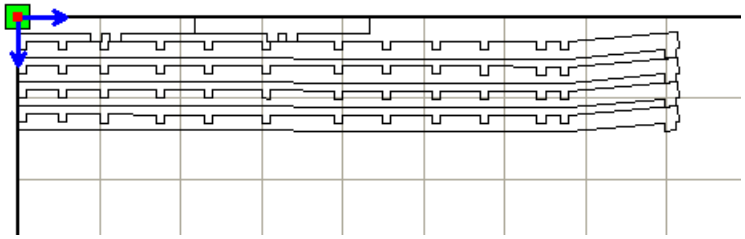
The tests were carried out for various mode including: Phugoid mode, Short Period Mode and the given graphs were observed. The effect of Dutch Roll, Roll Damping and Spiral Mode were also generated in XPlane.



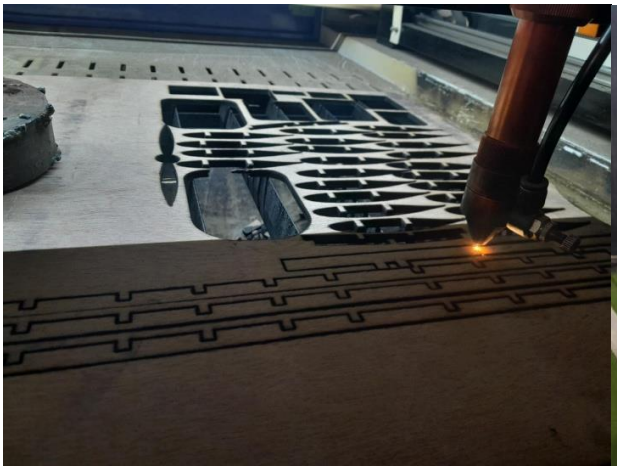
Manufacturing Process

The manufacturing process involves cutting the materials using either laser cutting or precision hand tools. As the laser cutting machine was already available in the manufacturing laboratory of IOE Pulchowk Campus, we decided to go with laser cutting method. The CAD parts drawn in CATIA were converted into dxf files and loaded into RDWorks software. The parts such as: ribs, longerons, frames, empennage which were designed in CAD were modified into rld files and at last uploaded to udisk as a rd file (the final form that could be uploaded into the laser cutting machine). The rd file was then uploaded to pendrive and inserted in the machine to load the file. On manual focus mode, the parts were loaded and cut. Safety steps as such as closing the lid of the machine while the machine was operating was followed. The parts were cut in three halves as to reduce the runtime of machine and avoid overheating. The first cut was made for ribs, fuselage frames along with a propeller. The second half included cutting of longerons wed and the tail while the final half included cutting of spars.





The cut parts were arranged in a proper order and joined with each other using Vega glue. Fabrication process included creation of the aircraft parts and joining them with the help of adhesive. The joining process indeed was tricky as we had to use our discretion to figure out which part would go where but overall; it was a fun task to complete. The following pictures include the laser cut parts and final aircraft assembly for the manufacturing laboratory.







Conclusion

The 1:14 scale Piper PA-38 Tomahawk prototype successfully replicates the full-scale aircraft's structural properties and was successfully fabricated.